

Artificial Intelligence (AI-2002)

Lecture 8: Informed (Heuristic) Search Strategies

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Informed (Heuristic) Search Strategies

- One that uses domain-specific hints about the location of goals.
- Can find solutions more efficiently than an uninformed strategy.
- The hints come in the form of a heuristic function, denoted $h(n)$:

$h(n)$ = estimated cost of the cheapest path from the state at node n to a goal state.

- For example, in route-finding problems, we can estimate the distance from the current state to a goal by computing the straight-line distance on the map between the two points.

Greedy best-first search

- Greedy best-first search is a form of best-first search that expands first the node with the Greedy best-first search lowest $h(n)$ value —the node that appears to be closest to the goal.
- This is likely to lead to a solution quickly.
- So the evaluation function $f(n) = h(n)$.

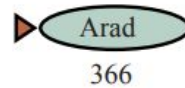
Greedy best-first search

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

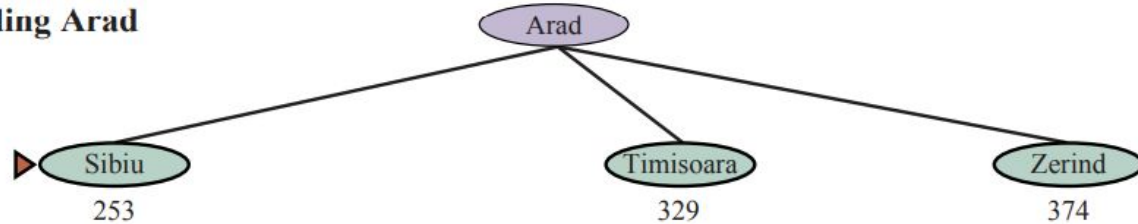
Figure 3.16 Values of h_{SLD} —straight-line distances to Bucharest.

Greedy best-first search

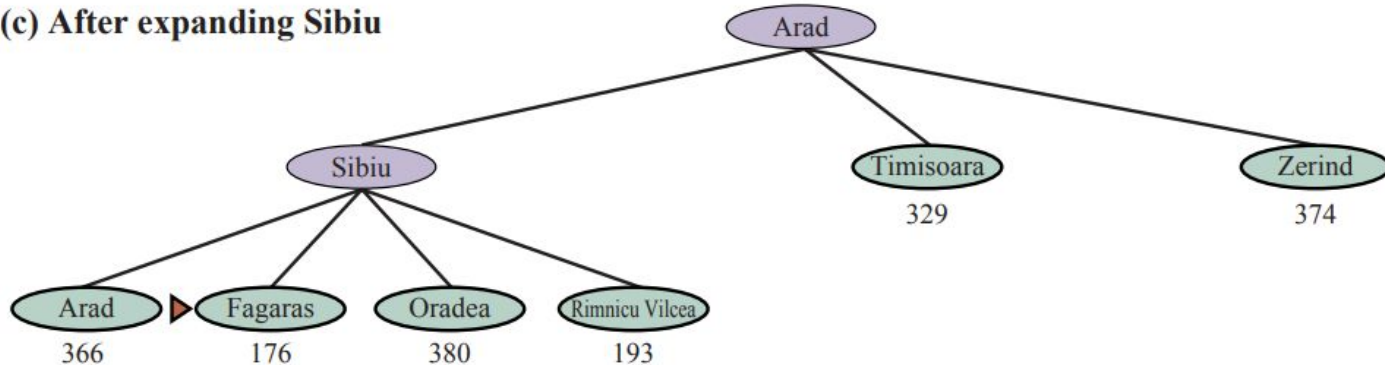
(a) The initial state



(b) After expanding Arad

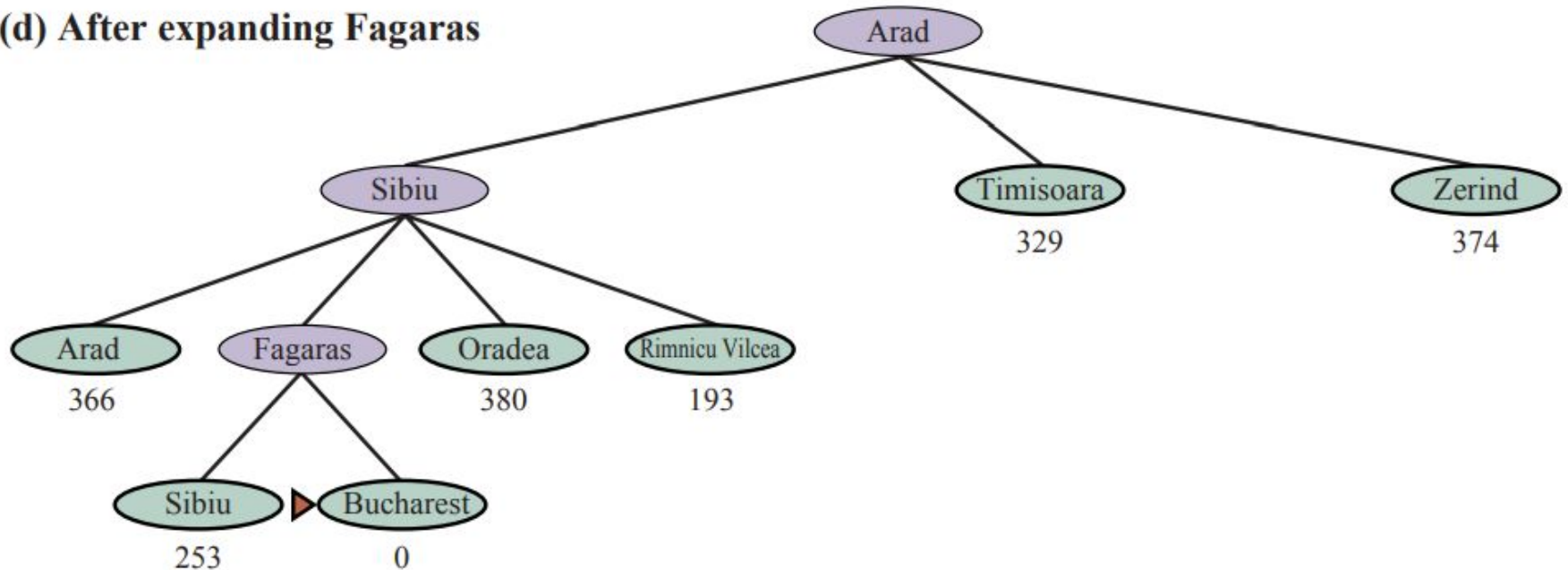


(c) After expanding Sibiu



Greedy best-first search

(d) After expanding Fagaras



Greedy best-first search

- This is why the algorithm is called “greedy”—on each iteration it tries to get as close to a goal as it can, but greediness can lead to worse results than being careful.
- Greedy best-first graph search is complete in finite state spaces, but not in infinite ones.
- The worst-case time and space complexity is $O(|V|)$.
- With a good heuristic function, however, the complexity can be reduced substantially, on certain problems reaching $O(bm)$.

A* search

- A* search uses the evaluation function:

$$f(n) = g(n) + h(n)$$

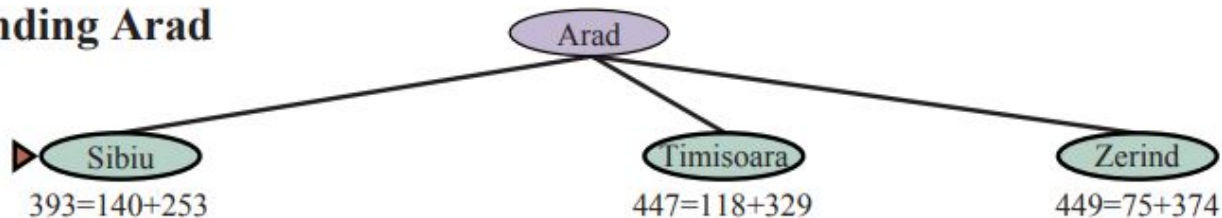
- where $g(n)$ is the path cost from the initial state to node n , and $h(n)$ is the estimated cost of the shortest path from n to a goal state.
- $f(n)$ = estimated cost of the best path that continues from n to a goal.

A* search

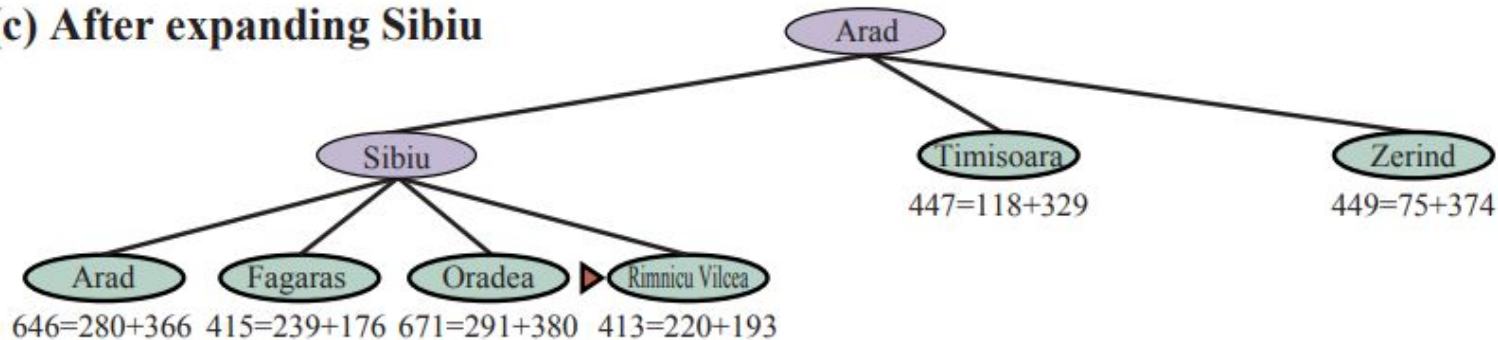
(a) The initial state



(b) After expanding Arad

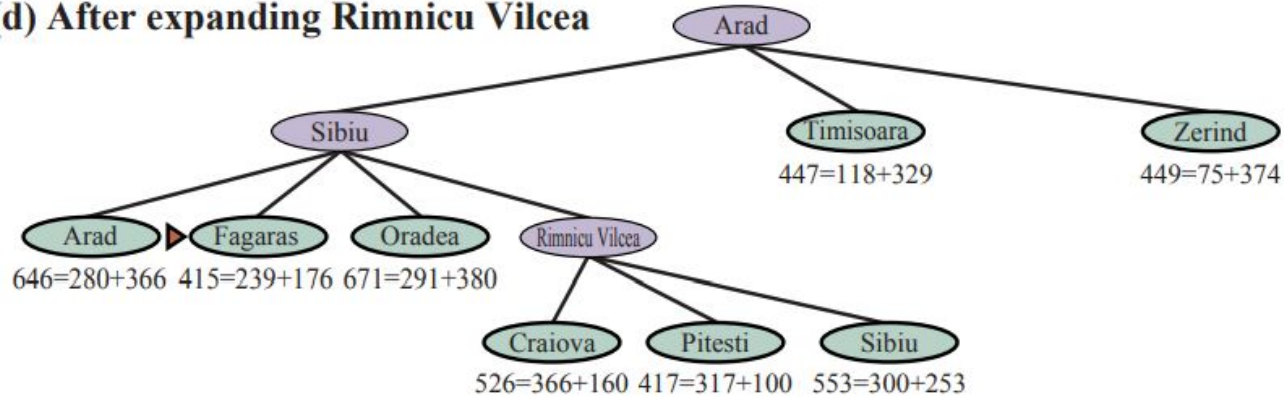


(c) After expanding Sibiu

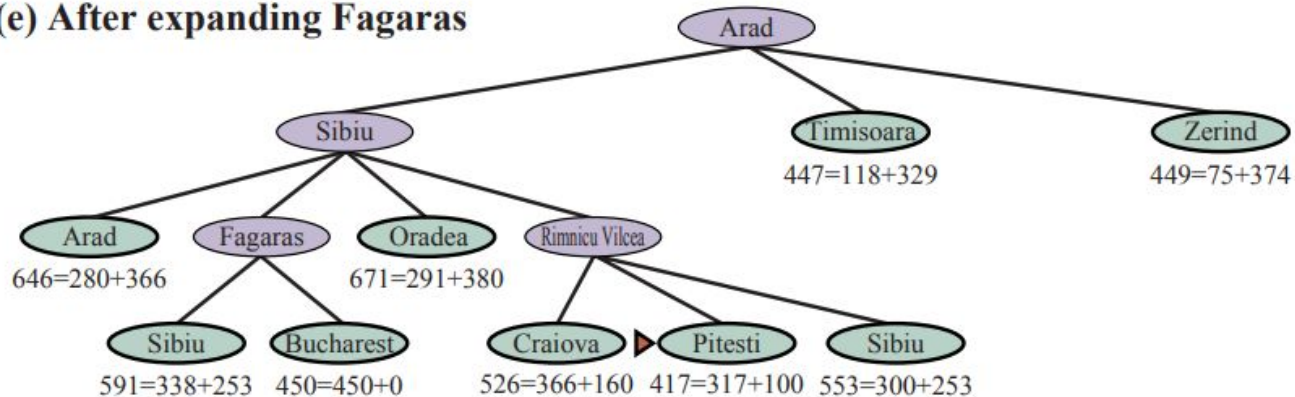


A* search

(d) After expanding Rimnicu Vilcea

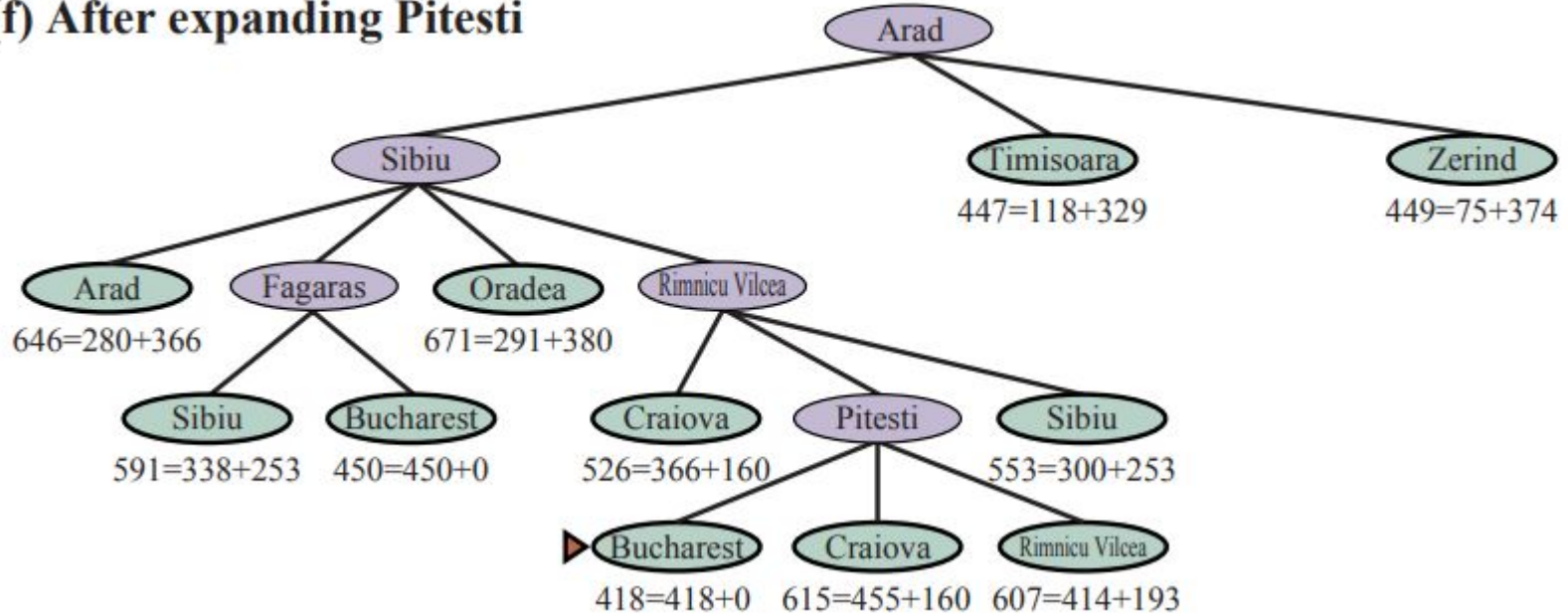


(e) After expanding Fagaras



A* search

(f) After expanding Pitesti



A* search

- A * search is complete.
- Whether A* is cost-optimal depends on certain properties of the heuristic.
- A key property is admissibility: an admissible heuristic is one that never overestimates the cost to reach a goal.

A* search

- A heuristic $h(n)$ is consistent if, for every node n and every successor n' of n generated by an action a , we have:

$$h(n) \leq c(n, a, n') + h(n')$$

- This is a form of the triangle inequality.
- Every consistent heuristic is admissible (but not vice versa).
- So with a consistent heuristic, A* is cost-optimal.

A* search

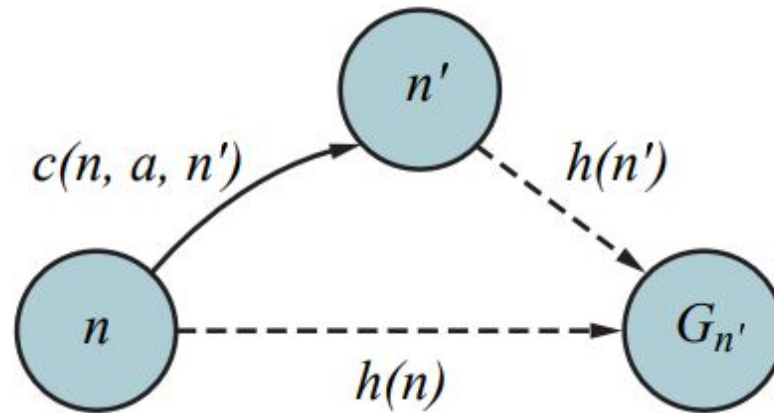


Figure 3.19 Triangle inequality: If the heuristic h is **consistent**, then the single number $h(n)$ will be less than the sum of the cost $c(n, a, a')$ of the action from n to n' plus the heuristic estimate $h(n')$.

A* search

